The Incentive to Ignore

Selective Attention and Audience Design in Distributed Discovery

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Abstract

A shared signal can be individually useful and socially harmful when many searchers turn the same evidence into the same action. This paper studies that tension in a finite exact model of distributed discovery. Each agent has an independent private clue and may use a common clue; discovery occurs when at least one action finds the target, while correct agents split a unit prize. The first shared-signal user adds a potentially superior route. Every duplicate user removes an independent route. For any finite team, the discovery optimum therefore assigns the shared clue to one role when its accuracy exceeds private accuracy, to no role when it is worse, and to zero or one role at equality. Private equal-split incentives need not implement that allocation: an exact finite census contains excessive use, all-attend equilibria, and welfare- relevant multiplicity. Binding audience design restores the one-reader count; within a registered symmetric garbling family, reducing precision does not repair over-publicity. A budget-balanced universal-pooling rule implements the optimal count under public access. Allowing every role to condition on both clues does not improve the planner optimum in the complete deterministic, agreement-respecting, label-equivariant disagreement class for three labels, although equilibrium wedges remain. A larger raw-policy audit supplies a counterexample to any unrestricted interpretation. All numerical statements come from checksum-validated exact runs. Experimental statements are predictions only: no participants were recruited, no human data were collected, and no experiment was conducted.

Keywords: selective attention; shared information; audience design; information avoidance; collective search; distributed discovery.

Evidence status. DD-C-0059, DD-C-0060, DD-C-0062–DD-C-0064, and DD-C-0066 are verified theorems. DD-C-0061, DD-C-0065, and DD-C-0067 are independently reproduced bounded exact classifications. DD-C-0068 is an independently reproduced negative scope result. Experimental contrasts are predictions, not observations.

Publication status. Working paper; not submitted, not peer reviewed, and no DOI.

1 Introduction

Organizations often treat information access as an unqualified good. A shared report is copied to the entire team, a dashboard is placed in every workflow, or the same retrieved context is supplied to every model agent. If the evidence is accurate, broad dissemination appears harmless. Yet information matters through the actions it induces. When several searchers use the same clue in the same way, they may concentrate on one candidate and abandon independent routes that would have covered other possibilities. The relevant organizational question is therefore not merely whether evidence is useful. It is who should use it, under what action rule, and with what incentives.

result	claim	evidence category	status
One-reader discovery	DD-C-0059	theorem	verified
Equal-split thresholds	DD-C-0060	theorem	verified
Reward census	DD-C-0061	bounded exact	independently reproduced
Audience and garbling	DD-C-0062–0063	theorems	verified
Universal pooling	DD-C-0064	theorem	verified
Voluntary audience	DD-C-0065	bounded exact	independently reproduced
Conditional planner	DD-C-0066	theorem	verified
Conditional equilibrium	DD-C-0067	bounded exact	independently reproduced
Raw-policy boundary	DD-C-0068	negative result	independently reproduced
Treatment effects	none	prediction	not observed

Table 1: Evidence status attaches to each statement, not to the paper as a whole.

Source runs: `20260721T212943Z_DD-012_9ed0928e_4a3f1ba62b`, `20260721T215811Z_DD-013_09c07448_cdac4fb512`, `20260721T222047Z_DD-014_f5f099a8_ea0276dd16`; claims: DD-C-0059–DD-C-0068; generator: `distributed_discovery.papers.build_incentive_to_ignore`; input SHA-256 prefixes: `81d6a760a71f`, `7514d4f97cd4`, `c355420c6b98`.

This paper isolates that question in a deliberately small discovery model. There is one target among three labels. Each of N agents observes an independent private clue with accuracy p. One common clue has accuracy q. An action directly names a label, and team discovery is the union event that at least one action names the target. The first role that follows the common clue may replace a weaker private route with a stronger route. A second follower does not create another common route: it repeats the first. The planner sees a first-use benefit and a duplicate-use loss.

The main theorem is correspondingly sharp. If nobody follows the shared clue, failure probability is $(1-p)^N$. If $k \geq 1$ roles follow it, failure probability is $(1-q)(1-p)^{N-k}$. The first-reader discovery margin is $(1-p)^{N-1}(q-p)$; every later margin is strictly negative. Thus exactly one reader is optimal when $q > p$, no reader when $q < p$, and zero or one at equality. The result is about union discovery inside a frozen follow/private action class. It is not a general theorem that public information should be hidden.

Private incentives introduce a second margin. A correct agent splits a unit prize with all other correct agents. A role selecting shared-following trades the probability of being right against competition from agents taking the same action and against the coverage externality imposed on the team. Exact payoff thresholds characterize weak and strict pure attention-count equilibria. All-ignore is stable exactly when $q \leq p$, so strategic ignorance cannot support complete non-use when one reader is uniquely socially optimal. The distortion instead appears mainly as duplicate attention: on the registered 175-cell grid, 63 cells have excessive-use equilibria and 24 admit all-attend although exactly one reader is optimal.

Audience design separates receipt from voluntary use. If an institution can bind which roles receive the shared clue, the same first-use argument chooses an audience of one when $q > p$. Sending a lower-precision version to more roles does not dominate that allocation in the registered symmetric garbling family. If everyone must retain access, a universal-pooling reward gives every role $1/N$ exactly when the team discovers. Each unilateral mode change then has the sign of the corresponding discovery change, implementing the optimal count without an external subsidy. That rule does not select reader identity, solve hidden-discovery problems, or prove robustness to richer policies.

The conditional-policy extension tests whether the value of role differentiation is an artifact of unconditional follow-or-ignore behavior. Every role observes both clues. When the clues agree, a deterministic label-equivariant policy takes the common label. When they disagree, it takes the private label, the public label, or the remaining third label. Those three rules are complete within the declared symmetry class. For $p \geq 1/3$, replacing a contrarian role with a private-dominant role weakly improves discovery; the one-public-role planner classification survives. Yet a separate two-

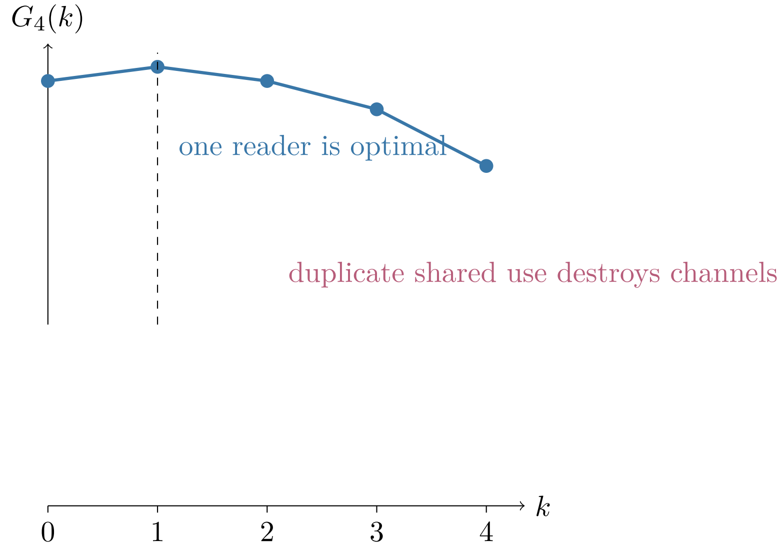

Figure 1: Restricted social discovery by the number k of shared-signal followers for $N = 4$, $p = 1/2$, and $q = 3/4$. First use raises discovery; every duplicate use lowers it.

Source runs: `20260721T212943Z_DD-012_9ed0928e_4a3f1ba62b`; claims: DD-C-0059, DD-C-0060, DD-C-0061; generator: `distributed_discovery.papers.build_incentive_to_ignore`; input SHA-256 prefixes: `b199943481f9`.

label audit of every raw deterministic table finds complementary constant policies with discovery one. That counterexample is not a nuisance to discard. It marks precisely where the restricted theorem stops.

The contribution is the exact discovery-portfolio mechanism linking first use, duplicate use, private reward, audience design, and conditional action rules. Strategic ignorance, information avoidance, rational inattention, partial publicity, herding, and contrarian behavior all have substantial prior literatures. The paper does not claim ownership of those general concepts or a verified novelty certificate. Its narrower contribution is to make one finite mechanism auditable from formulas, complete rational grids, independent enumerators, corruption tests, immutable run IDs, and public data.

The paper proceeds from literature and model to social discovery, private equilibrium, audience design, implementation, conditional policies, institutional implications, and experimental predictions. Appendices give proofs, evidence categories, finite-grid details, and reproduction information.

2 Related literature

2.1 Strategic ignorance and information avoidance

Strategic ignorance is an established economic object. Agents may avoid information because remaining uninformed changes later incentives, commitment, or responsibility. Carrillo and Mariotti [2000] analyze ignorance as a self-disciplining device. The broad information-avoidance literature spans instrumental, hedonic, and strategic motives and documents that people do not always demand useful information [Golman et al., 2017]. Our mechanism differs in two respects. First, non-use is an ex-ante action mode in a team discovery game, not a claim about psychological avoidance. Second, the social value comes from preserving an independent action route, not from protecting a single decision maker from temptation or affect.

Those differences matter for language. “Ignore” here means that a role does not use the common clue as its action guide. Under binding access it literally means non-delivery; under universal access it means private-dominant action. The model neither observes nor predicts whether a person reads, remembers, or emotionally avoids information. It shows how the same behavior can have a positive discovery externality in one portfolio technology.

2.2 Endogenous and rational attention

Rational-inattention models make information processing an endogenous choice under explicit costs or capacity constraints [Sims, 2003]. That literature typically optimizes an information structure or allocation of attention over states and actions. The present model freezes two signals and three simple action modes. There is no entropy cost and no continuous attention budget. The advantage is exact finite accounting: receipt, use, action correlation, and coverage can be separated without identifying a reduced-form capacity cost.

The distinction also protects the result from an overbroad inference. The one-reader theorem does not say that an organization should employ only one attentive person. Every nonreader still uses a private clue. The theorem says that one common clue should produce at most one common-following action when the objective is union discovery and private clues remain available. Total evidence access and effective action-channel count are different quantities.

2.3 Public information and partial dissemination

Public information can improve individual decisions yet interact adversely with coordination motives. Morris and Shin [2002] study the social value of public information in a coordination environment. Cornand and Heinemann [2008] analyze partial dissemination as a way to moderate the effects of publicity. Those papers establish that audience size can be a policy variable; their objectives, signal structures, and strategic actions differ from union discovery. Here publicity is costly because multiple users collapse distinct search routes into the same route even without a desire to coordinate.

This paper’s audience theorem is consequently neither a restatement nor a rebuttal of the coordination literature. It is an exact result for a coverage technology. The model makes the information boundary operational: a binding audience changes who can choose public-following; a voluntary audience changes only the available deviation set; a recommendation leaves access intact; and an action assignment constrains behavior after information arrives.

2.4 Herding and contrarian experimentation

Social-learning and experimentation models study when agents repeat observed choices, free-ride on information production, or take contrarian actions that generate new evidence. Smith et al. [2021] connect informational herding, optimal experimentation, and contrarianism in a dynamic setting. Our third-option policy is superficially contrarian but structurally different: it chooses the unused third label when private and shared clues disagree, within a one-shot model with no learning from outcomes. It creates an alternative action channel, not future public evidence.

The raw-policy audit reinforces this caution. When agreement-respecting and label-equivariance restrictions are removed in a two-label fixture, two roles can choose complementary constants and cover both labels with certainty. Such territorial behavior resembles role assignment more than inference. It can be socially powerful while being epistemically unresponsive. A theory of efficient contrarianism must therefore say whether actions are required to respect evidence, whether labels are symmetric, and whether fixed territorial coverage is admissible.

2.5 Collective search and portfolio design

Distributed discovery treats actions as a portfolio whose union matters. The same evidence can be valuable for accuracy and harmful for coverage if it correlates actions. That portfolio view connects attention to task allocation, organizational design, and multi-agent system architecture. Yet a public clue can also improve action quality, reduce wasted searches, or facilitate later coordination. Those objectives are recorded separately rather than folded into a single composite score.

The companion Information Sharing Frontier makes the same aggregation-versus- rescue tradeoff explicit for point, shortlist, exclusion, and strategic binary channels [Nakajima, 2026]. The present paper owns selective attention and audience design; the companion owns signal geometry, recovery budgets, and its selected-equilibrium sharing result. Organizational exploration and transparency remain broader adjacent mechanisms [March, 1991, Lazer and Friedman, 2007, Bernstein, 2012].

Boundary. Related literatures already study strategic ignorance, information avoidance, endogenous attention, partial publicity, herding, and contrarian experimentation. The evidence here supports only the declared exact discovery models and bounded comparisons; it supplies no verified novelty claim.

3 Model

3.1 Signals and actions

There are $N \geq 2$ agents and a target $\theta \in \{1, 2, 3\}$ drawn uniformly. Each agent i has a private clue X_i. A shared clue Y is common across roles. Conditional on the target, private clues are mutually independent and independent of Y. Their laws are

$$\mathbb{P}(X_i = \theta \mid \theta) = p, \qquad \mathbb{P}(X_i = x \neq \theta \mid \theta) = \frac{1-p}{2},$$

and likewise with accuracy q for Y. Each agent chooses one label. Team discovery is

$$D = \mathbf{1}\{\exists i : a_i = \theta\}, \qquad G = \mathbb{E}[D].$$

Action quality is the expected fraction of correct actions. Action diversity is the expected number of distinct labels. These three metrics are reported separately because a design can raise one and lower another.

3.2 Attention modes

In DD-012, each role makes an ex-ante choice between private-following, $a_i = X_i$, and shared-following, $a_i = Y$. We call the latter role a reader or attender as shorthand for use. The count k is the number of shared-following roles. The model is anonymous: every profile with the same k has the same social discovery and type payoffs.

If at least one agent is correct, a unit prize is divided equally among correct agents. Thus agent i's realized reward is

$$u_i = \frac{\mathbf{1}\{a_i = \theta\}}{\sum_j \mathbf{1}\{a_j = \theta\}},$$

with zero assigned when nobody is correct. Exact payoff accounting implies $\sum_i \mathbb{E}[u_i] = G$. A weak pure attention-count equilibrium permits no strict gain from changing one's mode; strict equilibrium requires every available unilateral change to lose strictly.

3.3 Audience and institutions

DD-013 adds an audience size m. Under binding delivery, exactly m roles receive the shared clue and follow it. Under voluntary use, recipients choose private- or shared-following; nonrecipients cannot choose the latter. Garbling replaces q by a symmetric delivered accuracy $g \leq q$ on a registered grid. The institution registry separately records delivery, recommendations, action assignments, observable outcomes, transfers, commitment, and subsidy.

3.4 Conditional policies

DD-014 gives every role both clues. Policies respect agreement: if $X_i = Y$, the common label is chosen. On disagreement, a deterministic label-equivariant policy chooses X_i, chooses Y, or chooses the remaining label. The role profile (a, b, c) counts private-dominant, public-dominant, and third-option contrarian policies. This is the complete disagreement class under the stated symmetry and agreement restrictions, not the unrestricted table class.

3.5 Evidence and scope

The analytic theorems hold at the scopes stated in their ledger records. Bounded censuses use exact rational grids: DD-012 and DD-013 cover $N = 2, \ldots, 8$ and $p, q \in \{1/3, 1/2, 2/3, 3/4, 5/6\}$; DD-014 covers $N \in \{2, 3, 4\}$ on the same accuracy grid. Every displayed number traces to an immutable run manifest and an independently implemented state enumerator. No human behavior is inferred.

4 First-use value and duplicate-use loss

When $k = 0$, every action follows an independent private clue. Failure requires all N private clues to miss:

$$G_N(0) = 1 - (1 - p)^N.$$

For $k \geq 1$, all shared-followers take the same action. Failure requires the shared clue to miss and all $N - k$ remaining private clues to miss:

$$G_N(k) = 1 - (1 - q)(1 - p)^{N-k}.$$

These identities are definitions plus conditional independence; they are not asymptotic approximations.

The first-use margin is

$$G_N(1) - G_N(0) = (1 - p)^{N-1}(q - p).$$

Its sign compares shared and private accuracy. For every later user,

$$G_N(k + 1) - G_N(k) = -(1 - q)p(1 - p)^{N-k-1} < 0$$

for interior accuracies. Once the shared route exists, another follower removes one private route without creating a new shared route.

N	planner k	weak equilibrium k	all-attend equilibrium?	exact discovery wedge
2	1	2	yes	1/8
3	1	2,3	yes	1/16
4	1	3	no	3/32
5	1	3	no	3/64
6	1	4	no	7/128
7	1	4	no	7/256
8	1	5	no	15/512

Figure 2: Planner and weak equal-split attention counts along the registered $p = 1/2, q = 3/4$ slice. The table separates multiplicity from the best-equilibrium discovery gap.

Source runs: 20260721T212943Z_DD-012_9ed0928e_4a3f1ba62b; claims: DD-C-0059, DD-C-0060, DD-C-0061; generator: distributed_discovery.papers.build_incentive_to_ignore; input SHA-256 prefixes: b199943481f9.

This proves the One-Reader Theorem (DD-C-0059): the optimum is uniquely $k = 1$ when $q > p$, uniquely $k = 0$ when $q < p$, and exactly $k \in \{0, 1\}$ when $q = p$. The proof works for every finite $N \geq 2$. It distinguishes an extensive margin from an intensive margin. The first use changes the route set; duplicate use changes only how many roles take the shared route.

The formula also clarifies scaling. The first-use gain is multiplied by $(1-p)^{N-1}$ because it matters only when the other private routes fail. In a large team with strong private clues, the incremental union value of any one route can be small. Duplicate loss likewise depends on remaining private routes. The optimal count remains one, but the welfare magnitude need not remain large.

Action quality tells a different story. A shared follower has standalone accuracy q; replacing a p-accurate private follower can raise the expected fraction correct when $q > p$. Multiple followers may therefore improve average quality while reducing union discovery. An institution optimizing a weighted combination could choose a different audience. The paper never calls the union optimum universally optimal across objectives.

5 Private attention equilibrium

The equal-split prize makes correctness and congestion jointly relevant. A shared follower is correct whenever $Y = \theta$ but then competes with every other shared follower and any private follower who also happens to be correct. A private follower has accuracy p and competes with a random number of correct roles. Exact binomial expectations give both type payoffs at every attention count. Their weighted sum equals discovery.

Let Δ_k be the gross payoff gain to a private follower who becomes the $(k + 1)$st shared follower. The proof record derives

$$\Delta_k = q(1-p)\mathbb{E}\left[\frac{1}{k+1+Z_k}\right] - p(1-q)\mathbb{E}\left[\frac{1}{1+Z_k}\right],$$

where Z_k is the number of correct actions among the appropriate remaining private roles. A Bernoulli coupling shows that Δ_k is strictly decreasing in k. The sequence completely characterizes weak and strict anonymous pure equilibria: entry at k must not gain and exit from k must not gain.

Two boundary conditions are especially interpretable. All-ignore is a weak equilibrium exactly when $q \leq p$. Therefore, when $q > p$, the decentralized game cannot support complete non-use even

$p\backslash q$	1/3	1/2	2/3	3/4	5/6
1/3	$\mathrm{E}_{0,1;0,1}$	$\mathrm{X}_{2;1}$	$\mathrm{X}_{3,4;1}$	$\mathrm{X}_{4;1}$	$\mathrm{X}_{4;1}$
1/2	$\mathrm{E}_{0;0}$	$\mathrm{E}_{0,1;0,1}$	$\mathrm{X}_{2;1}$	$\mathrm{X}_{3;1}$	$\mathrm{X}_{4;1}$
2/3	$\mathrm{E}_{0;0}$	$\mathrm{E}_{0;0}$	$\mathrm{E}_{0,1;0,1}$	$\mathrm{E}_{1;1}$	$\mathrm{X}_{3;1}$
3/4	$\mathrm{E}_{0;0}$	$\mathrm{E}_{0;0}$	$\mathrm{E}_{0;0}$	$\mathrm{E}_{0,1;0,1}$	$\mathrm{X}_{2;1}$
5/6	$\mathrm{E}_{0;0}$	$\mathrm{E}_{0;0}$	$\mathrm{E}_{0;0}$	$\mathrm{E}_{0;0}$	$\mathrm{E}_{0,1;0,1}$

Figure 3: Attention-wedge phase map at $N = 4$. Each cell reports $\text{category}_{\text{weak equilibrium counts;planner counts}}$. E is efficient; X is excessive shared-signal use; M records equilibrium multiplicity with a discovery difference.

Source runs: `20260721T212943Z_DD-012_9ed0928e_4a3f1ba62b`; claims: DD-C-0060, DD-C-0061; generator: `distributed_discovery.papers.build_incentive_to_ignore`; input SHA-256 prefixes: `838acdc5d7ac`.

though ignorance may be strategically valuable elsewhere. All-attend is weak exactly when

$$q \geq \frac{Np}{1 + (N-1)p}.$$

This threshold can hold while the unique social optimum remains one reader. Shared evidence can be sufficiently attractive privately that every role uses it despite the social loss from duplicate action.

The registered exact census records 1,050 attention profiles across 175 cells. It contains 63 excessive-attention cells, three with welfare-relevant equilibrium multiplicity, and 24 in which all-attend is an equilibrium although one reader is uniquely optimal. These are bounded classifications (DD-C-0061), not frequency estimates for organizations. They show existence and structure inside the grid.

6 The attention wedge

The attention wedge is the difference between planner discovery and the best discovery achieved by a weak equilibrium under a specified reward rule. A zero wedge can arise because equilibrium implements the planner count, because equilibrium is multiple and includes an optimum, or because several counts tie in discovery. Those cases should not be conflated. The phase map records full sets rather than selecting a favorable equilibrium silently.

Three mechanisms generate divergence. First, a role does not internalize the private route it removes by becoming another shared follower. Second, equal prize splitting makes a highly accurate common action privately attractive even when it is redundant. Third, identity symmetry can leave multiple agents willing to occupy the single socially valuable reader role. These are distinct from a cognitive processing cost, a signaling motive, or a desire to coordinate.

The model also identifies an important non-result. Complete ignorance does not explain the excessive-use region: all-ignore cannot be stable when $q > p$. The incentive to ignore is marginal and role-specific. A well-designed portfolio asks most roles to ignore the common action recommendation while one role uses it. Calling that pattern "organizational ignorance" would obscure the active private search performed by nonreaders.

audience	discovery	quality	distinct actions	voluntary weak use	binding optimum?
0	15/16	1/2	295/128	0	no
1	31/32	9/16	571/256	1	yes
2	15/16	5/8	125/64	2	no
3	7/8	11/16	25/16	3	no
4	3/4	3/4	1	3	no

Figure 4: Binding audience frontier and voluntary use for $N = 4$, $p = 1/2$, $q = 3/4$. Audience, use, and action assignment are distinct institutional variables.

Source runs: `20260721T215811Z_DD-013_09c07448_cdac4fb512`; claims: DD-C-0062, DD-C-0065; generator: `distributed_discovery.papers.build_incentive_to_ignore`; input SHA-256 prefixes: `357e7b32c1dc`.

Reward interventions can alter the wedge. Sole-rescue and marginal-coverage rewards pay for action contribution rather than duplicated correctness. Universal pooling divides the team discovery prize equally among all roles. A public-reader license binds the permitted count. In the registered census, sole-rescue, marginal-coverage, universal pooling, and the binding license match the social-optimum count correspondence in every cell. Off-shared-success instead produces unique all-ignore in every cell, and an assigned-reader subsidy fails to leave any socially optimal weak equilibrium. Negative mechanism results are preserved alongside successful ones.

These rules differ in observability. Marginal coverage requires knowing whether a role uniquely rescues discovery. Pooling requires team discovery but not the identity of the correct role. A license requires access control and commitment. The numerical correspondence alone does not establish implementation under hidden outcomes, collusion, limited liability, or endogenous participation.

7 Audience design

Binding delivery turns the attention count into an institutional choice. With audience $m = 0$, failure is $(1 - p)^N$. With any $m \geq 1$, failure is $(1 - q)(1 - p)^{N-m}$. The same first-use and duplicate-use differences therefore yield the Binding Audience Theorem (DD-C-0062): full-precision delivery to one recipient is optimal when $q > p$, no delivery when $q < p$, and zero or one at equality; every audience of two or more is strictly worse.

The theorem concerns access paired with prescribed following. A recipient may instead retain the option to act on a private clue. DD-013 enumerates every voluntary profile for each audience. Across 1,050 audience settings and 4,025 profiles, voluntary use differs from binding use in 656 settings, permits excessive use in 273, and exhibits welfare-relevant multiplicity in eight. All 70 insufficient-use settings have audience zero, so the grid contains no positive audience with equilibrium use below the global binding optimum. This last count is a bounded observation, not a general theorem that recipients never under-attend.

Exclusive delivery has a simple implementation advantage: only one role can take the common action. It also creates governance demands. The institution must select a recipient, prevent leakage, and ensure that excluded roles retain independent evidence. Rotating the recipient can distribute responsibility across repeated tasks, but the one-shot theorem contains no repeated-game incentive. Random assignment selects identity ex ante without changing the optimal count.

delivered accuracy g	audience	discovery	binding optimum	relation
1/3	1	11/12	31/32	strictly below
1/3	2	5/6	31/32	strictly below
1/3	4	1/3	31/32	strictly below
1/2	1	15/16	31/32	strictly below
1/2	2	7/8	31/32	strictly below
1/2	4	1/2	31/32	strictly below
2/3	1	23/24	31/32	strictly below
2/3	2	11/12	31/32	strictly below
2/3	4	2/3	31/32	strictly below
3/4	1	31/32	31/32	ties
3/4	2	15/16	31/32	strictly below
3/4	4	3/4	31/32	strictly below

Figure 5: Precision versus publicity on a registered slice. Garbling does not repair over-publicity within the declared symmetric family; full precision to one recipient weakly dominates every listed design.

Source runs: `20260721T215811Z_DD-013_09c07448_cdac4fb512`; claims: DD-C-0062, DD-C-0063; generator: `distributed_discovery.papers.build_incentive_to_ignore`; input SHA-256 prefixes: `357e7b32c1dc`.

7.1 Precision versus publicity

Reducing signal precision is sometimes proposed as an alternative to limiting audience. The registered garbling family replaces accuracy q by $g \in \{1/3, 1/2, 2/3, 3/4, 5/6\}$ with $1/3 \le g \le q$, then sends the garbled signal to a positive audience. Its failure probability is $(1-g)(1-p)^{N-m}$. When $q > p$, this is weakly above the one-recipient full-precision failure probability, with equality only at $g = q$ and $m = 1$. At equality $q = p$, no delivery adds a tie; when $q < p$, any positive use is strictly harmful in the frozen action class.

Thus the full-precision binding optimum weakly dominates all 2,625 registered garbling rows. Exactly 105 rows tie, each a one-recipient, full-precision design with $q \ge p$; the remaining 2,520 are strictly worse (DD-C-0063). This is not a Blackwell-order theorem over arbitrary experiments. A richer garbling could change actions differently, and a different objective could value broad noisy coordination. The result only shows that symmetric precision reduction does not repair duplicate-route loss here.

8 Information firewalls and implementation

An information firewall is not a single mechanism. The DD-013 registry records eight institutional forms spanning binding delivery, voluntary receipt, recommendations, public action assignment, garbling, licenses, and rewards. For each, it states who observes the shared clue, which action choices remain, what outcome is observable, whether transfers occur, what commitment is needed, and whether an external subsidy is assumed.

Binding exclusive delivery implements the audience count by construction. It may be administratively natural when tasks can be partitioned before evidence arrives. A recommendation is weaker: if all roles retain access and equal-split rewards, the voluntary equilibrium can duplicate the public action. An action assignment can force role differentiation but needs enforceability. A payment rule can change incentives without restricting receipt, but may require outcome verification.

Universal pooling is unusually transparent. Give every role $1/N$ exactly when the team discovers

rule/institution	implemented or weak counts	evidence	boundary
assigned reader reward	4	exact	DD-012 reward/access rule
equal split	3	exact	DD-012 reward/access rule
marginal coverage	1	exact	DD-012 reward/access rule
off shared success	0	exact	DD-012 reward/access rule
public reader license	1	exact	DD-012 reward/access rule
sole rescue	1	exact	DD-012 reward/access rule
universal pooling	1	exact	DD-012 reward/access rule
binding exclusive delivery	1	exact	DD-013 access assignment
public universal pooling	1	exact	DD-013 ex-post balanced public rule

Figure 6: Mechanism comparison on $N = 4$, $p = 1/2$, $q = 3/4$. Identical count outcomes can rely on different access, observability, commitment, and budget assumptions.

Source runs: `20260721T212943Z_DD-012_9ed0928e_4a3f1ba62b`, `20260721T215811Z_DD-013_09c07448_cdac4fb512`; claims: DD-C-0061, DD-C-0064; generator: `distributed_discovery.papers.build_incentive_to_ignore`; input SHA-256 prefixes: `838acdc5d7ac`, `357e7b32c1dc`.

and zero otherwise. Each role's expected payoff is $G_N(k)/N$. Consequently every unilateral mode change has exactly the sign of the corresponding discovery change. The weak equilibrium count correspondence equals the binding optimum for every finite $N \geq 2$ and interior p, q, and is strict when $p \neq q$. Realized payments sum to one on discovery and zero on failure, so the rule is ex-post budget balanced against the unit discovery prize and requires no external subsidy (DD-C-0064).

Pooling implements a count, not an identity. When $q > p$, any one of N roles can be the reader. An assignment or rotation rule is still needed if identity matters. Pooling also assumes that team discovery is observable and contractible, that all roles accept equal ex-post payoffs, and that side transfers or collusion do not change incentives. Role aversion, career concerns, or heterogeneous costs are outside the proof.

9 Conditional attention

The unconditional model may seem behaviorally brittle. A person who sees both clues need not follow one source in every state. DD-014 therefore lets every role condition on (X_i, Y). Agreement must select the common label. Under determinism and label-equivariance, disagreements form one orbit and exactly three outputs survive: the private label, the public label, or the remaining third label. This proves completeness inside the declared class.

For a profile (a, b, c), condition on whether the shared clue is correct. If there is no public-dominant role, failure probability is

$$q(1-p)^{a+c} + (1-q)(1-p)^a \left(\frac{1+p}{2}\right)^c.$$

If $b \geq 1$, failure probability is

$$(1-q)(1-p)^a \left(\frac{1+p}{2}\right)^c.$$

When $p \geq 1/3$, replacing a contrarian with a private-dominant role weakly reduces failure because $1 - p \leq (1+p)/2$, strictly when $p > 1/3$. The planner can therefore remove contrarians without loss and recover the unconditional zero-or-one-public-role classification (DD-C-0066).

At $p = 1/3$, a private clue is uninformative and contrarian roles can tie private- dominant roles. They never improve the optimum. This explains why the 75-cell census reports contrarian use in an optimal profile in exactly 15 cells but no positive conditional gain over the unconditional class. The conditional action space changes the set of tied implementations, not the best discovery value.

$p\backslash q$	1/3	1/2	2/3	3/4	5/6
1/3	$\mathrm{E}_{004/013/103/112/202/211/301/310/400}$	$\mathrm{W}_{013/112/211/310}$	$\mathrm{W}_{013/112/211/310}$	$\mathrm{W}_{013/112/211/310}$	$\mathrm{W}_{013/112/211/310}$
1/2	E_{400}	$\mathrm{E}_{310/400}$	W_{310}	W_{310}	W_{310}
2/3	E_{400}	E_{400}	$\mathrm{E}_{310/400}$	E_{310}	W_{310}
3/4	E_{400}	E_{400}	E_{400}	$\mathrm{E}_{310/400}$	W_{310}
5/6	E_{400}	E_{400}	E_{400}	E_{400}	$\mathrm{E}_{310/400}$

Figure 7: Conditional-policy phase map at $N = 4$. Planner profiles are triples (private-dominant, public-dominant, contrarian); W marks a positive best-equilibrium wedge and E marks zero. Long entries record exact planner ties at uninformative private accuracy.

Source runs: `20260721T222047Z_DD-014_f5f099a8_ea0276dd16`; claims: DD-C-0066, DD-C-0067, DD-C-0068; generator: `distributed_discovery.papers.build_incentive_to_ignore`; input SHA-256 prefixes: `7a0bffdd95bd`.

Private equilibrium remains nontrivial. Across 775 anonymous profiles, the exact census contains 115 weak and 53 strict equilibria. In 49 cells, every weak equilibrium is planner-optimal. In 23, even the best weak-equilibrium discovery is below the planner optimum. Conditional policies can eliminate a wedge in some regions but do not eliminate it generally on the grid (DD-C-0067).

9.1 The unrestricted audit

The symmetry theorem must not be described as an unrestricted policy theorem. DD-014 separately audits two agents and two labels at four rational accuracy cells. A raw deterministic policy maps four private/shared observation pairs to binary actions, yielding sixteen tables and 256 ordered two-role profiles per cell. Both exact evaluators enumerate all 1,024 profiles and 16,384 labeled states.

Complementary constant policies cover both labels with certainty, so raw planner discovery is one in every audit cell. The embedded private/public class falls short by 1/16, 1/12, or 1/9. Some raw equilibria also reach values outside the restricted class. This negative result (DD-C-0068) exposes the role of agreement-respect and evidence responsiveness. It does not solve unrestricted $M = 3$ conditional attention, but it decisively prevents promotion of the restricted theorem.

10 Institutional implications

The analysis suggests a sequence of design questions rather than a universal prescription. First, identify the objective. Union discovery rewards coverage; average accuracy rewards correctness; coordination rewards correlated action. Second, identify effective action channels. The number of people receiving a report can exceed the number of distinct routes produced. Third, distinguish access, use, and assignment. A firewall changes feasible information; a recommendation changes advice; a role assignment changes action rights. Fourth, match incentives to the observable outcome.

For a research team, one role might be assigned to investigate the leading shared hypothesis while others retain independent hypotheses. For an investment process, one analyst might own the consensus case while others test orthogonal failure modes. For a multi-agent system, one agent might use retrieved common context and others use independently sampled or source-separated context. These are analogies, not validated field interventions. Real use would require task dependence, costs, leakage, skill heterogeneity, and error consequences.

The results favor explicit role differentiation over vague exhortations to “think independently.” A binding audience makes the information boundary clear. Universal pooling makes the discovery objective common. A license makes the scarce common-following role explicit. Yet each solution creates governance questions: who receives the role, how rotation works, how private evidence is protected, how discovery is verified, and how failure responsibility is shared.

Precision control is not a substitute for audience control in the registered family. Sending a noisier signal widely still correlates actions and sacrifices private routes. Likewise, conditional policies are not a complete substitute for institutional allocation. Private-dominant behavior preserves dispersion, but equal-split incentives can still overuse public-dominant roles. The policy space and the reward space interact.

The raw audit adds a countervailing warning. If fixed territorial actions are allowed, complete coverage may be achieved without informative evidence. That can be useful for exhaustive search but dangerous when actions are costly, labels are numerous, or correctness matters beyond discovery. Institutional design should record whether actions must respond to evidence rather than assuming all diversity is epistemically meaningful.

11 Experimental predictions

The exact models generate falsifiable directional predictions for a future synthetic or human study, but they do not validate those predictions empirically. A broadcast treatment should generate more shared-signal use than a one-reader assignment under equal-split rewards. When $q > p$, one-reader access should produce higher union discovery than full broadcast in the frozen task. A nonbinding recommendation to ignore should fail to reproduce a binding license when private incentives favor duplicate use. Universal pooling should shift attention counts toward the discovery optimum if participants understand and trust the payoff rule.

Conditional-policy treatments distinguish receipt from response. Subjects or agents can be asked to choose an ex-ante rule before signals are realized, or their realized actions can be classified after both clues appear. The theorem predicts no planner gain from third-option contrarian roles for $p \geq 1/3$ in the symmetric task, though uninformative-private cells permit ties. The bounded equilibrium census predicts that conditional choice reduces some but not every attention wedge.

An experiment must also measure leakage and compliance. A designated nonreader who remembers the public clue is not equivalent to a role denied access. A recipient who sees the clue but chooses a private action is not equivalent to a recipient forced to follow it. Outcome variables should therefore include receipt, recall, chosen rule, realized action, correctness, union discovery, distinct actions, and payoffs. Treatment contrasts must state which information boundary they manipulate.

The existing DD-011 package is synthetic-only. It registers hypotheses, estimands, randomization, response scenarios, and power calculations under declared models. Any attention extension must retain calibration failures and distinguish rational equilibrium, noisy best response, salience, partial compliance, memory leakage, role aversion, and social preferences. Power under those scenarios is conditional design evidence, not evidence about humans.

Ethics gate. No participants were recruited. No human data were collected. No experiment was conducted. Human deployment would require a separate ethics, consent, privacy, retention, compensation, and institutional review process.

12 Limitations

The target is uniform and finite. Wrong signals are symmetric. Private clues are homogeneous and conditionally independent. There is one shared clue, one action per role, and one-shot timing. Signal acquisition is free inside the attention model. Discovery is a binary union event. Correct roles split

contrast	outcomes	directional prediction	evidence label
Broadcast versus one reader	public-signal use; discovery	broadcast increases use; one-reader access raises discovery when $q > p$	prediction from DD-C-0059/DD-C-0062
Recommendation versus license	attention count; compliance	nonbinding advice need not implement the optimum; binding access does	institutional prediction
Equal split versus pooling	attention count; payoff	pooling aligns unilateral count changes with discovery	theorem DD-C-0064
Conditional recommendation	policy category; discovery	conditional policies do not universally remove the wedge	bounded prediction DD-C-0067

Figure 8: Experimental treatment map. These are model-derived predictions and registered contrasts, not observations. No participants were recruited and no human experiment was conducted.

Source runs: `20260721T212943Z_DD-012_9ed0928e_4a3f1ba62b,20260721T215811Z_DD-013_09c07448_cdac4fb512,20260721T222047Z_DD-014_f5f099a8_ea0276dd16,20260721T185647Z_DD-011_fa0271d9_fcaa647c55`; claims: DD-C-0056, DD-C-0059, DD-C-0062, DD-C-0064, DD-C-0067; generator: `distributed_discovery.papers.build_incentive_to_ignore`; input SHA-256 prefixes: `81d6a760a71f, 7514d4f97cd4, c355420c6b98, 0d586f4001c8`.

a unit prize, and the principal can sometimes observe discovery or enforce access. Each assumption supports exactness and limits external interpretation.

Correlated private errors could make nominally private routes redundant. Heterogeneous accuracy could justify assigning shared use to a weak private role even when identities matter. Multiple shared sources could create a source- selection problem. Reusable evidence could make broad dissemination valuable across future tasks. Action costs could favor concentration. False-positive harms could make exhaustive coverage undesirable. Sequential observation could turn an action into public evidence and introduce experimentation incentives.

The equilibrium analysis is pure and ex ante. It does not characterize mixed strategies, learning, bounded rationality, ambiguity, or communication. Equal prize splitting is only one reward technology. The successful pooling rule relies on observable team discovery and common acceptance of equal shares. Institutional costs are not subtracted from welfare.

The audience theorem restricts actions to follow-shared or follow-private. The garbling theorem covers a finite symmetric accuracy family, not arbitrary information design. The conditional theorem covers deterministic, agreement- respecting, label-equivariant policies. The raw audit shows that relaxing those conditions can change the optimum drastically. No unrestricted $M = 3$ policy claim is made.

Grid counts are exact for their registrations but are not probabilities over real environments. The figures are selected slices, not empirical response surfaces. The literature review establishes adjacency and prior art but does not certify novelty. The paper is a working paper: it has no DOI, is not submitted, and is not peer reviewed.

13 Conclusion

Shared evidence has two different margins in distributed discovery. Its first use can add a better route; duplicate use can erase independent routes. In the frozen finite model, that distinction yields a one-reader discovery theorem, an exact private-attention equilibrium characterization, and bounded evidence of excessive common-signal use. Audience design turns the optimal count into an access decision. Symmetric garbling does not repair over-publicity, while universal pooling implements the optimal count under its observability and commitment assumptions.

Allowing joint conditioning on private and shared clues does not overturn the planner result inside the complete declared three-label symmetry class. Contrarian roles tie only when private information is uninformative, and the equilibrium wedge persists in part of the registered grid. The larger raw audit shows why that scope statement matters: unrestricted territorial policies can achieve full coverage for reasons unrelated to inference.

The practical lesson is not to suppress information reflexively. It is to design an action portfolio deliberately. Receipt, use, allocation, rewards, precision, and observability are separate levers. The evidence supports exact statements about one discovery technology and offers predictions for future tests. It does not establish human behavior or a universal rule for public information.

A Proof of the one-reader result

For completeness, fix any target θ. With no shared follower, conditional independence gives

$$\mathbb{P}(D = 0 \mid \theta) = \prod_{i=1}^{N} \mathbb{P}(X_i \neq \theta \mid \theta) = (1-p)^N.$$

The expression is target-independent, so averaging over the uniform prior does not change it. With $k \geq 1$, every shared follower fails exactly when $Y \neq \theta$. Conditional on that event, the $N-k$ private followers fail independently, giving $(1-q)(1-p)^{N-k}$. Subtracting adjacent discovery values produces the margins in Section 4. Interior p, q make every duplicate margin strictly negative. This proves DD-C-0059.

The proof uses neither the equal-split prize nor voluntary choice. It is a social identity for a binding count. Strategic equilibrium enters only after type payoffs are defined. This separation is useful for verification: the primary closed form and the direct target/signal enumerator must agree before any payoff or mechanism claim is considered.

B Equal-split payoff accounting

Let $K = \sum_i \mathbf{1}\{a_i = \theta\}$ be the number of correct actions. Realized rewards sum to one when $K > 0$ and zero otherwise. Therefore

$$\sum_i \mathbb{E}[u_i] = \mathbb{E}[\mathbf{1}\{K > 0\}] = G.$$

This identity is checked for every profile. Within an anonymous count, all roles of a type have the same expected payoff. Multiplying the type payoff by its count and summing must reproduce discovery exactly.

The independent verifier does not call the primary payoff formula. It loops over target, shared signal, and every action-relevant private signal state, constructs actions, identifies winners, and divides the prize. It reconstructs unilateral deviations and equilibrium sets from these direct payoffs. Altered discovery, reward, and equilibrium fields are rejected.

C Conditional-policy completeness and failure formulas

The symmetric group on three labels acts transitively on agreement pairs and on ordered disagreement pairs. Agreement-respect fixes the output on the diagonal. For a representative disagreement (x, y), the stabilizer leaves three equivariant choices: x, y, or the remaining label. Applying relabelings generates exactly the private-dominant, public-dominant, and contrarian tables. No fourth deterministic label-equivariant disagreement output exists.

When $Y = \theta$, a public-dominant role succeeds surely, while private-dominant and contrarian roles succeed exactly when their private clue is correct. When $Y \neq \theta$, a private-dominant role succeeds with probability p, a public- dominant role never succeeds, and a contrarian succeeds when its private clue equals the label other than Y and θ, with probability $(1-p)/2$. Multiplying independent failure terms gives the formulas in Section 9.

For $p \geq 1/3$,

$$1 - p \leq \frac{1+p}{2}.$$

Replacing a contrarian factor by a private-dominant factor therefore weakly reduces both relevant failure expressions. The inequality is strict for $p > 1/3$. Setting $c = 0$ then embeds the unconditional class, completing the planner proof for DD-C-0066.

D Registered finite evidence

DD-012 evaluates $N = 2$ through 8 and the five-by-five rational accuracy grid. Its 175 cells contain 1,050 attention profiles and 6,300 reward/profile rows. The separate verifier traverses 3,319,200 labeled states. DD-013 evaluates 1,050 binding audience rows, 4,025 voluntary profiles, 2,625 feasible garbling rows, and 175 mechanism cells using another 3,319,200-state reproduction. DD-014 evaluates 775 anonymous conditional profiles across 75 cells, verifies 346,275 main labeled states, and separately audits 1,024 raw profiles and 16,384 states.

Exact fractions are authoritative. Displayed decimals, where used only for plotting coordinates, are derived from those fractions and are not separate claims. Every run has a clean source commit, configuration hash, dependency-lock hash, input hashes, output hashes, UTC timestamps, command, environment record, stdout, stderr, validation summary, and corruption record. Passing runs are not rerun to refresh timestamps.

E Evidence classification and predictions

A definition states a model object. An identity follows directly from those definitions. A theorem has a human-readable proof and the checks required by the claim-status policy. A bounded computational result reports only its frozen grid. Independent reproduction uses a materially separate evaluator. A negative result preserves a failed generalization or counterexample. An experimental prediction is implied by a model but unobserved.

The distinction matters for this paper's strongest prose. "One reader is optimal" is a theorem only inside the follow/private union-discovery class. "Sixty-three cells exhibit excessive use" is an exact bounded count, not a population prevalence. "Broadcast will increase use" is a prediction, not an empirical fact. "Contrarians never help" would be false without the stated symmetry boundary, as the raw audit demonstrates.

F Reproducibility record

The paper generator validates every source file against the output checksum in its immutable run manifest before generating a table or figure. Generated assets contain claim IDs, run IDs, generator name, and input checksum prefixes. The bibliography is copied from the validated project bibliography; all cited keys and claim IDs are resolved. Tectonic compiles twice with a fixed source epoch. The two PDF byte strings must match. The build rejects unresolved references, unresolved citations, and overfull boxes.

The committed validation record contains the PDF checksum, page count, source runs, generated asset names, and all input checksums. Poppler renders every page to an image for visual review. The public site copies the PDF only when its checksum matches validation, then publishes that checksum alongside the working- paper boundary. This workflow supports reproducibility; it does not confer peer review, archival status, or a DOI.

G Institutional specification checklist

An application should specify the following objects before borrowing a result from this paper. The checklist is part of the evidence boundary, not an implementation theorem.

1. **Target and loss.** State the action universe, the target event, and whether the objective is union discovery, average accuracy, false-positive control, cost-adjusted value, or a multi-objective vector. A one-reader result for union coverage does not rank designs under another loss.

2. **Evidence law.** Record private and shared accuracies, dependence, heterogeneity, source overlap, and whether evidence persists across tasks. Nominally separate retrieval calls do not guarantee independent errors.

3. **Information boundary.** Distinguish non-delivery, delivery with a recommendation, delivery with a binding action rule, delayed delivery, and memory leakage. A recipient set is not automatically a user set.

4. **Action map.** State whether roles follow a clue, condition on multiple clues, choose an assigned territory, communicate, or observe prior actions. The DD-014 raw audit shows that this boundary can dominate results.

5. **Payoff and observability.** Specify who is paid, what success is observable, whether marginal contribution can be verified, whether transfers balance ex post, and whether participation or limited liability binds.

6. **Identity rule.** If exactly one common-signal role is desirable, state how identity is selected, rotated, or contested. Count implementation does not settle distributive fairness or role burden.

7. **Leakage and compliance.** Measure receipt, recall, declared mode, realized action, and deviations separately. An information firewall with unmeasured leakage cannot be interpreted as binding access.

8. **Evidence category.** Label a theorem, exact grid result, simulation, prediction, or observation accurately. Never use a synthetic power result as a completed experiment or a grid frequency as an empirical prevalence.

For a multi-agent implementation, the checklist can be encoded in a task manifest. Each agent should have declared sources, context visibility, action rights, and reward outputs. A validation harness should reconstruct effective action channels, not merely count messages or contexts. Logs should retain the mapping from evidence receipt to action so that correlation induced by common context is observable.

For a human study, additional ethics and governance gates are mandatory. The task must avoid deceptive responsibility shifting, disclose compensation, protect private reasoning traces, define retention and deletion, and separate research participation from employment evaluation. Nothing in the present synthetic package satisfies those institutional requirements automatically.

H Open extensions and falsification targets

The most direct theoretical extension introduces heterogeneous private accuracies p_i. A plausible conjecture is that, when exactly one public role is desirable, it should be assigned to a role with relatively weak private accuracy. That statement is not proved here: equal-split payoffs and identity-specific externalities must be recalculated. A counterexample could arise if a high-skill role uses shared evidence more effectively under a richer action map.

Correlated private clues are another priority. Positive dependence reduces the effective number of private routes and can change the first-use magnitude. The zero-or-one structure may survive for a single duplicated public route under some exchangeable laws, but the simple factorization into $(1-p)^N$ does not. An extension needs a pinned joint distribution and an independent enumerator; pairwise correlation alone is insufficient to identify union failure.

Multiple shared sources create a portfolio problem. The planner may want one user per shared source, several users for a particularly strong source, or no user when sources overlap. Source choice and audience design then interact. The relevant state is a source-action incidence structure rather than one attention count. DiscoveryBench tasks should expose that structure explicitly instead of scoring only total information access.

Dynamic attention changes both objectives and incentives. A contrarian action can reveal information for later roles, and visible successes can induce herding. Stopping-time discovery and fixed-budget discovery are different objectives. Optional DD-015 remains registered rather than executed in this program because the required static evidence, paper, benchmark, and synthetic-design queue took priority. No dynamic theorem is inferred from the static conditional audit.

Several observations would falsify a literal behavioral interpretation of the registered models. Widespread private-dominant action under very high q could indicate role aversion, mistrust, or misunderstanding. Persistent public- dominant use under a binding nonreceipt treatment would reveal leakage or a failed manipulation. Improved discovery from broad broadcast could indicate that private actions were already correlated, that recipients condition in a richer way, or that communication creates new routes. Each outcome would call for a revised model rather than post-hoc relabeling of the original theorem.

The institutional predictions also invite direct mechanism tests. Compare a nonbinding recommendation, a binding license, universal pooling, and a marginal- coverage reward while holding signal realization fixed. Record whether the rules change attention counts, action diversity, and discovery as predicted. Retain comprehension and calibration failures. A null or reversed result would be evidence about the behavioral or implementation assumptions, not a numerical error in the exact finite calculations.

Finally, external validity requires domain-specific stakes. Search actions can be cheap hypotheses, costly experiments, irreversible investments, or harmful interventions. Union discovery is compelling only when false positives and action costs are controlled. A deployed system should use a registered multi-objective evaluation and report tradeoffs rather than promote the one-reader count as a universal organizational rule.

References

Ethan S. Bernstein. The transparency paradox: A role for privacy in organizational learning and operational control. *Administrative Science Quarterly*, 57(2):181–216, 2012. doi: 10.1177/0001839212453028.

Juan D. Carrillo and Thomas Mariotti. Strategic ignorance as a self-disciplining device. *The Review of Economic Studies*, 67(3):529–544, 2000. doi: 10.1111/1467-937X.00142.

Camille Cornand and Frank Heinemann. Optimal degree of public information dissemination. *The Economic Journal*, 118(528):718–742, 2008. doi: 10.1111/j.1468-0297.2008.02139.x.

Russell Golman, David Hagmann, and George Loewenstein. Information avoidance. *Journal of Economic Literature*, 55(1):96–135, 2017. doi: 10.1257/jel.20151245.

David Lazer and Allan Friedman. The network structure of exploration and exploitation. *Administrative Science Quarterly*, 52(4):667–694, 2007. doi: 10.2189/asqu.52.4.667.

James G. March. Exploration and exploitation in organizational learning. *Organization Science*, 2 (1):71–87, 1991. doi: 10.1287/orsc.2.1.71.

Stephen Morris and Hyun Song Shin. Social value of public information. *American Economic Review*, 92(5):1521–1534, 2002. doi: 10.1257/000282802762024610.

Yohei Nakajima. When does information sharing improve decentralized discovery? Working paper, 2026. URL `https://yoheinakajima.github.io/distributed-discovery/publications/information-sharing-frontier.html`.

Christopher A. Sims. Implications of rational inattention. *Journal of Monetary Economics*, 50(3): 665–690, 2003. doi: 10.1016/S0304-3932(03)00029-1.

Lones Smith, Peter Norman Sørensen, and Jianrong Tian. Informational herding, optimal experimentation, and contrarianism. *The Review of Economic Studies*, 88(5):2527–2554, 2021. doi: 10.1093/restud/rdab001.

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The paper's numerical evidence comes from three clean immutable runs: '20260721T212943Z_DD-012_9ed0928e_4a3f1ba62b', '20260721T215811Z_DD-013_09c07448_cdac4fb512', and '20260721T222047Z_DD-014_f5f099a8_ea0276dd16'. The experimental map also references the synthetic-only DD-011 design run. Each generated figure or table prints its contributing run IDs, claim IDs, generator, and input checksum prefixes. Full SHA-256 values appear in 'generated/provenance.json' and the run manifests.

The authoritative evidence labels are in 'claims/claims.yml'. Theorems are supported by human-readable proofs and exact checks; bounded classifications are limited to their registered grids; the raw-policy result is a negative scope audit; and treatment statements are unobserved predictions. No participant, organization, or deployed multi-agent system was studied.

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